

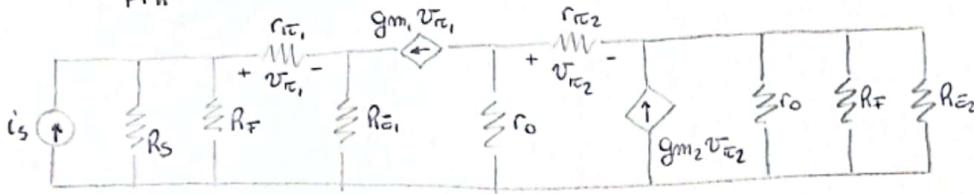
$$\textcircled{1} \quad V_{CC} = R_P(I_{C3} + 2I_{B3}) - V_{BE} \rightarrow I_{B3} = \frac{V_{CC} + V_{BE}}{R_P(2 + \beta_{FP})} = 7 \mu A \quad I_{C3} = \beta_F I_{B3} = 1 \text{ mA}$$

$$I_{RE2} = \frac{V_{CC} - V_O}{R_{E2}} = 0.85 \text{ mA} \quad \left\{ \begin{array}{l} I_{C3} = I_{C1} - I_{B2} \\ I_{RE2} = I_{C2} + I_{B2} + I_{B1} \end{array} \right. \rightarrow \left\{ \begin{array}{l} I_{C1} = I_{C3} + I_{B2} \\ I_{RE2} = I_{B2}(1 + \beta_{FP}) + \frac{I_{C3} + I_{B2}}{\beta_{FN}} \end{array} \right.$$

$$I_{B2} = \frac{I_{RE2} - I_{C3}/\beta_{FN}}{1 + \beta_{FP} + 1/\beta_{FN}} = 5.6 \mu A \quad I_{C1} = I_{C3} + I_{B2} \approx I_{C3} = 1 \text{ mA} \quad I_{C2} = \beta_{FP} I_{B2} = 0.84 \text{ mA}$$

$$I_{B1} = \frac{I_{C1}}{\beta_{FN}} = 5 \mu A \quad V_{CE1} = V_O + V_{BEp} - R_{E1}(I_{B1} + I_{C1}) = 2.62 \text{ V}$$

$$R_F = \frac{-V_{CE2} - V_{BEp} - R_{E1}I_{E1}}{I_{B1}} = 524 \text{ k}\Omega$$



$$g_{m1} = \frac{I_{C1}}{V_T} = 38.46 \text{ mS}$$

$$r_{\pi1} = \frac{\beta_{FN}}{g_{m1}} = 5.2 \text{ k}\Omega$$

$$g_{m2} = 32.31 \text{ mS} \quad r_{\pi2} = 4.64 \text{ k}\Omega$$

$$A_1 = \frac{R_{out2}(1 + \beta_{O1})}{r_{\pi2} + R_{out2}(1 + \beta_{OP})} = 0.99$$

$$R_{out2} = r_{o1} \parallel R_F \parallel R_{E2} = 4.58 \text{ k}\Omega$$

$$\beta = -\frac{1}{R_F}$$

$$A_2 = -\frac{\beta_{ON} R_{im2} \parallel r_{o2}}{r_{\pi1} + R_{E1}(1 + \beta_{ON})} = -281.5$$

$$A_{OL} = R_s \parallel R_F \parallel R_{im1}, \quad A_1 A_2 = -2.566 \cdot 10^6$$

$$R_{im}^{OL} = R_s \parallel R_F \parallel R_{im1} = 9.18 \text{ k}\Omega$$

$$R_{out}^{OL} = r_{o1} \parallel R_F \parallel R_{E2} \parallel \frac{r_{\pi2} + r_{o2}}{1 + \beta_{OP}} = 1.33 \text{ k}\Omega$$

$$A_F = \frac{A_{OL}}{1 + \beta A_{OL}} = -453.15 \cdot 10^3$$

$$A_V = A_F \frac{1}{R_s} = -45.31$$

$$R_{im}^F = \frac{R_{im}^{OL}}{1 + \beta A_{OL}} = 1.56 \text{ k}\Omega$$

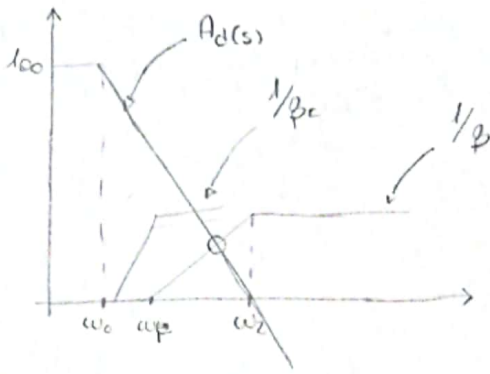
$$R_{im} = \frac{R_s \cdot R_{im}^F}{R_s - R_{im}^F} = 1.84 \text{ k}\Omega$$

$$R_{out} = R_{out}^F = \frac{R_{out}^{OL}}{1 + \beta A_{OL}} = 226.47 \text{ }\Omega$$

$$R_s^{\infty} = R_s + R_F \parallel R_{im1} = 121.65 \text{ k}\Omega$$

$$f_L = \frac{1}{2\pi R_s^{\infty} C_s} = 1.308 \text{ Hz}$$

$$(2) \quad A_v(s) = \frac{R_2}{R_1 + \frac{1}{sC_1}} = \frac{sC_1 R_2}{1 + sC_1 R_1}$$



$$\omega_p = \frac{1}{\beta_2} \omega_{cr} = \frac{1}{\beta_1} \omega_z = A_{d0} \omega_0$$

$$\omega_{cr} = \frac{\omega_z + \omega_p}{2} = 507.5 \text{ rad/s}$$

$$PH = 180^\circ - \arctan \frac{\omega_{cr}}{\omega_0} - \arctan \frac{\omega_{cr}}{\omega_p} + \arctan \frac{\omega_{cr}}{\omega_z} = 28.6^\circ$$

$$B = \frac{\omega_{cr}}{2\pi} = 80.77 \text{ kHz}$$

obtain $PH = 60^\circ$

solution #1

$$\omega_z = \frac{\omega_{cr}}{4} \quad \beta_{\infty} = \frac{R_1}{R_2} \rightarrow \frac{1}{\beta_{\infty}} = \frac{R_2}{R_1} = 68$$

$$\frac{1}{\beta_{\infty}} \omega_{cr} = A_{d0} \omega_0 \rightarrow \omega_{cr} = A_{d0} \omega_0 \beta_{\infty} = 14.71 \cdot 10^6$$

$$\omega_z = \frac{\omega_{cr}}{4} = \frac{1}{C_1^c R_1} \rightarrow C_1^c = \frac{4}{\omega_{cr} R_1} = 272 \text{ nF} \rightarrow \omega_p = \frac{1}{C_1^c (R_1 + R_2)} = 53.28 \text{ rad/s}$$

$$PH = 180^\circ - \arctan \frac{\omega_{cr}}{\omega_0} - \arctan \frac{\omega_{cr}}{\omega_p} + \arctan 4 = 75.96^\circ$$

solution #2

$$\tan 60^\circ = 1.73 \rightarrow \omega_z = \frac{\omega_{cr}}{2} \rightarrow C_1' = \frac{2}{\omega_{cr} R_1} = 136 \text{ nF} \rightarrow \omega_p = \frac{1}{C_1' (R_1 + R_2)} = 106.56 \text{ rad/s}$$

$$PH = 180^\circ - \arctan \frac{\omega_{cr}}{\omega_0} - \arctan \frac{\omega_{cr}}{\omega_p} + \arctan \frac{\omega_{cr}}{\omega_z} = 63.44 \text{ rad/s}$$

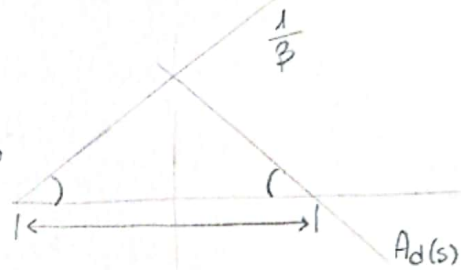
$$\beta = \frac{Z_1}{Z_1 + Z_2} = \frac{R_1 + \frac{1}{sC_1}}{R_1 + \frac{1}{sC_1} + R_2} = \frac{1 + sC_1 R_1}{1 + sC_1 (R_1 + R_2)}$$

$$\omega_z = \frac{1}{C_1 R_1} = 10^6 \text{ rad/s} \quad \omega_p = \frac{1}{C_1 (R_1 + R_2)} \approx 15 \text{ rad/s}$$

$$1 \cdot \omega_p = \beta_1 \omega_z \rightarrow \beta_1 = \frac{\omega_z}{\omega_p} = 66.67$$

$$A_{d0} \omega_0 = \frac{1}{\beta_1} \omega_{cr} \rightarrow \omega_{cr} = A_{d0} \omega_0 \beta_1 = 66.67 \cdot 10^6 \text{ rad/s}$$

$$PH = 180^\circ - \arctan \frac{\omega_{cr}}{\omega_0} - \arctan \frac{\omega_{cr}}{\omega_p} + \arctan \frac{\omega_{cr}}{\omega_z} =$$



$\frac{1}{\beta}$

$A_d(s)$